# STREAMLINE Replication Data Evaluation Report: 2025-12-23 12:18:23.069449

## General Pipeline Settings:

CV Partitions: 3
Partition Method: Stratified
Categorical Cutoff: 10
Statistical Significance Cutoff: 0.05
Random Seed: 42
Run From Notebook: False

## Feature Importance/Selection Settings:

Use Mutual Information: True
Use MultiSURF: False
Use TURF: False
TURF Cutoff: 0.5
MultiSURF Instance Subset: 470
Max Features to Keep: 5000
Filter Poor Features: True
Top Features to Display: 40
Export Feature Importance Plot: True

## ML Modeling Algorithms:

Artificial Neural Network: False
Category Gradient Boosting: False
Decision Tree: True
Elastic Net: False
ExSTraCS: False
Extreme Gradient Boosting: True
Genetic Programming: False
Gradient Boosting: False
K-Nearest Neighbors: True
Light Gradient Boosting: False
Logistic Regression: True
Naive Bayes: False
Random Forest: True
Support Vector Machine: False
XCS: False
eLCS: False

## Modeling Settings:

Primary Metric: balanced_accuracy
Training Subsample for KNN,ANN,SVM,and XGB: 0
Uniform Feature Importance Estimation (Models): True
Hyperparameter Sweep Number of Trials: 100
Hyperparameter Timeout: 900
Export Hyperparameter Sweep Plots: False

## LCS Settings (eLCS,XCS,ExSTraCS):

Do LCS Hyperparameter Sweep: False
nu: 1
Training Iterations: 2000
N (Rule Population Size): 2000
LCS Hyperparameter Sweep Timeout: 1200

## Stats and Figure Settings:

Export ROC Plot: True
Export PRC Plot: True
Export Metric Boxplots: True
Export Feature Importance Boxplots: True
Metric Weighting Composite FI Plots: balanced_accuracy
Top Model Features To Display: 40
Export Feature Correlations: True

## EDA and Processing Settings:

Engineering Missingness Cutoff: 1.0
Cleaning Missingness Cutoff: 1.0
Correlation Removal Threshold: 1.0
List of Exploratory Analysis Ran: ['Univariate Analysis', 'Feature Correlation']
List of Exploratory Plots Saved: []
Use Data Scaling: False
Use Data Imputation: False
Use Multivariate Imputation: False
Overwrite CV Datasets: True

## Target Training Dataset:

SNP_merged_train

## Applied to Following Replication Dataset(s):

D1 = SNP_merged_test

## Target Data Settings:

Data Path: ./data/SNPData
Output Path: ./Abdoul_Output
Experiment Name: Abdoul_thesis
Class Label: label
Instance Label: sample_id
Match Label: None
Ignored Features: None
Specified Categorical Features: None
Specified Quantitative Features: None

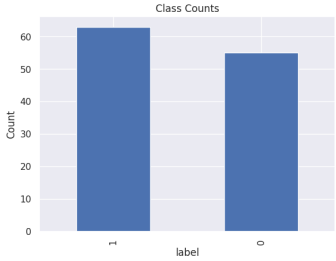
# Dataset and Model Prediction Summary: D1 = SNP_merged_test

## Data Processing/Counts Summary

| | Instances | Total Features | Categorical Features | Quantitative Features | Missing Values | Missing Percent | Class 0 | Class 1 |
|---|---|---|---|---|---|---|---|---|
| Original | 118 | 56717 | 56717 | 0 | 0 | 0.0 | 63 | 55 |
| C1 | 118 | 56717 | 56717 | 0 | 470 | 0.0001 | 55 | 63 |
| R1 | 118 | 67259 | 67259 | 0 | 470 | 0.0001 | 55 | 63 |
| Processed | 118 | 67259 | 67259 | 0 | 470 | 0.0001 | 55 | 63 |

**Cleaning (C) and Replication (R) Elements**
* C1 - Remove instances with no outcome
* R1 - Add/remove same features as Phase 1

## Class Balance (Processed)

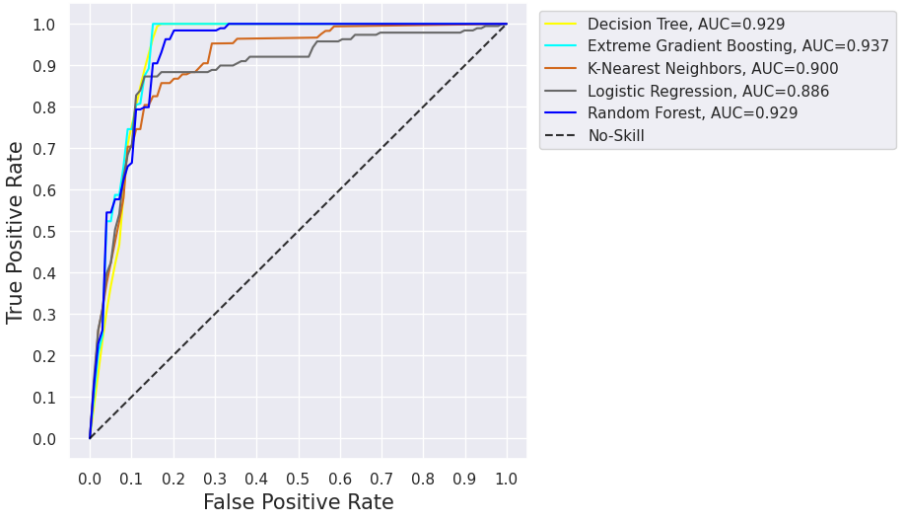


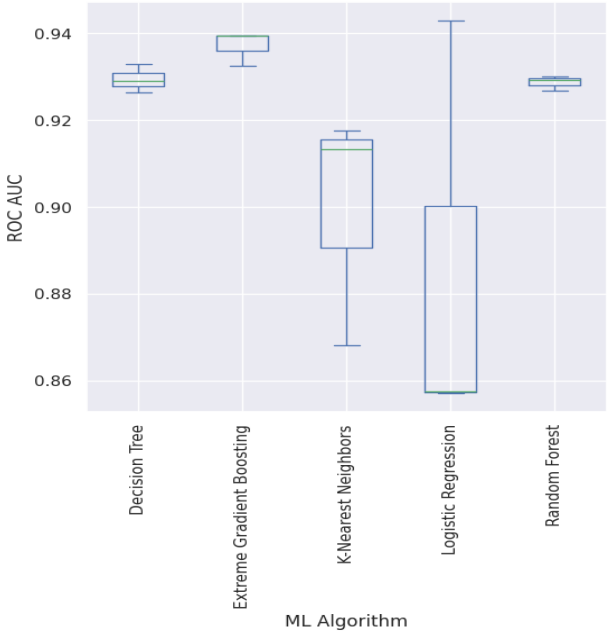

## Feature Correlations (Pearson)

No Feature Correlation Plot

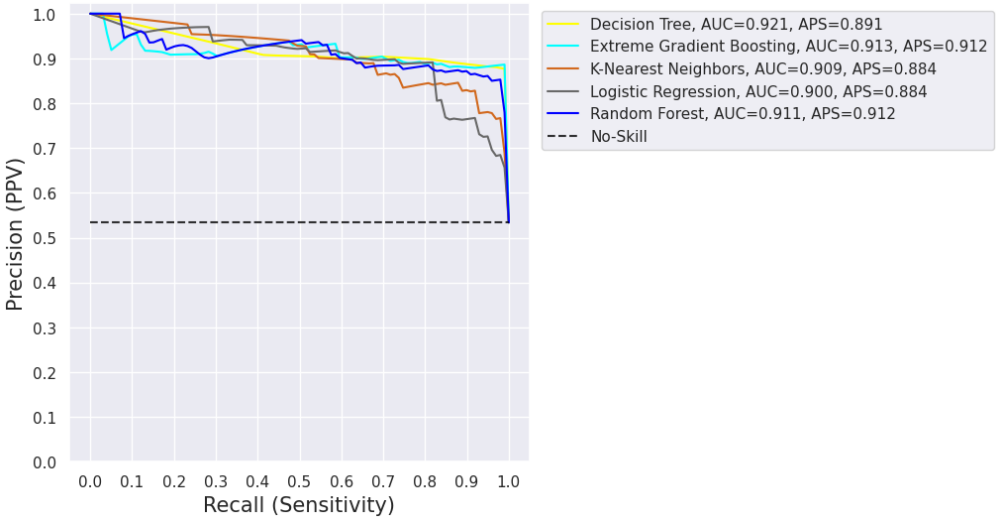
| Top ML Algorithm Results (Averaged Over CV Runs): |
|---|
| Best (ROC_AUC): Extreme Gradient Boosting = 0.937 |
| Best (Balanced Acc.): Decision Tree = 0.916 |
| Best (F1 Score): Decision Tree = 0.930 |
| Best (PRC AUC): Decision Tree = 0.921 |
| Best (PRC APS): Extreme Gradient Boosting = 0.912 |

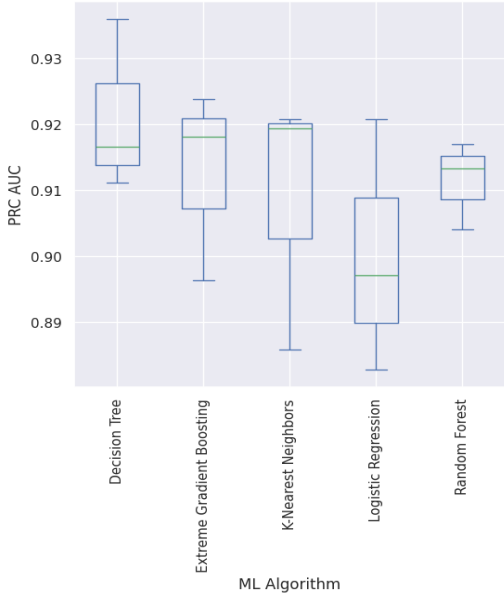
## ROC




## PRC




*Generated with STREAMLINE (0.3.4): (https://github.com/UrbsLab/STREAMLINE)*

# Average Model Prediction Statistics (Rounded to 3 Decimal Points)

## D1 = SNP_merged_test

| ML Algorithm | Balanced Accuracy | Accuracy | F1 Score | Sensitivity (Recall) | Specificity | Precision (PPV) | TP | TN | FP | FN | NPV | LR+ | LR- | ROC AUC | PRC AUC | PRC APS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Decision Tree | 0.916 | 0.921 | 0.93 | 0.989 | 0.842 | 0.878 | 62.333 | 46.333 | 8.667 | 0.667 | 0.986 | 6.293 | 0.012 | 0.929 | 0.921 | 0.891 |
| Extreme Gradient Boosting | 0.852 | 0.85 | 0.849 | 0.82 | 0.885 | 0.893 | 51.667 | 48.667 | 6.333 | 11.333 | 0.831 | 7.392 | 0.198 | 0.937 | 0.913 | 0.912 |
| K-Nearest Neighbors | 0.843 | 0.845 | 0.854 | 0.868 | 0.818 | 0.843 | 54.667 | 45.0 | 10.0 | 8.333 | 0.853 | 4.931 | 0.166 | 0.9 | 0.909 | 0.884 |
| Logistic Regression | 0.782 | 0.791 | 0.827 | 0.921 | 0.642 | 0.752 | 58.0 | 35.333 | 19.667 | 5.0 | 0.865 | 3.373 | 0.155 | 0.886 | 0.9 | 0.884 |
| Random Forest | 0.869 | 0.87 | 0.879 | 0.889 | 0.848 | 0.87 | 56.0 | 46.667 | 8.333 | 7.0 | 0.872 | 5.873 | 0.131 | 0.929 | 0.911 | 0.912 |

# Median Model Prediction Statistics (Rounded to 3 Decimal Points)

## D1 = SNP_merged_test

| ML Algorithm | Balanced Accuracy | Accuracy | F1 Score | Sensitivity (Recall) | Specificity | Precision (PPV) | TP | TN | FP | FN | NPV | LR+ | LR- | ROC AUC | PRC AUC | PRC APS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Decision Tree | 0.918 | 0.924 | 0.933 | 1.0 | 0.836 | 0.875 | 63.0 | 46.0 | 9.0 | 0.0 | 1.0 | 6.111 | 0.0 | 0.929 | 0.917 | 0.891 |
| Extreme Gradient Boosting | 0.835 | 0.831 | 0.828 | 0.762 | 0.909 | 0.898 | 48.0 | 50.0 | 5.0 | 15.0 | 0.769 | 7.683 | 0.262 | 0.939 | 0.918 | 0.917 |
| K-Nearest Neighbors | 0.871 | 0.873 | 0.884 | 0.905 | 0.836 | 0.864 | 57.0 | 46.0 | 9.0 | 6.0 | 0.885 | 5.529 | 0.114 | 0.913 | 0.919 | 0.883 |
| Logistic Regression | 0.769 | 0.78 | 0.817 | 0.921 | 0.618 | 0.734 | 58.0 | 34.0 | 21.0 | 5.0 | 0.872 | 2.411 | 0.128 | 0.857 | 0.897 | 0.883 |
| Random Forest | 0.872 | 0.873 | 0.882 | 0.889 | 0.855 | 0.869 | 56.0 | 47.0 | 8.0 | 7.0 | 0.87 | 5.784 | 0.13 | 0.929 | 0.913 | 0.912 |